An Integral-Shift Criterion for Dyadic Tail Recurrences

Erdős Problem #251, with the principal statements checked in Lean 4

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Abstract

Call a sequence $(T_N)_{N\geq 0}$ of rational numbers a *dyadic tail recurrence with digits* g when $T_{N+1} = 2T_N - g_{N+1}$ for every N, the digits g_N being arbitrary integers. For any such sequence we prove the block identity $T_{N+h} = 2^h T_N - B_{h,N}$, with $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$, and hence that the shift $T_{N+h} - T_N$ is an integer *if and only if* $(2^h - 1)T_N$ is; that integrality of one shift propagates to every later index; and, by Euler's congruence, that a term whose reduced denominator d is odd has an integral shift of length $\varphi(d)$, where φ denotes Euler's totient function. Consequently every such rational sequence has a fixed $h \geq 1$ whose shifts are integers from some index onwards. The contrapositive over the reals is an irrationality criterion: if a real sequence obeys the same recurrence with integer digits and, for every $h \geq 1$, the shift $T_{N+h} - T_N$ fails to be an integer for arbitrarily large N, then T_0 is irrational. The digits enter all of these statements only through the fact that they are integers.

Let $p_0 = 2, p_1 = 3, \ldots$ enumerate the primes and let $g_n = p_{n+1} - p_n$. We formalise in Lean both the exact finite identity

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}},$$

and, conditional only on summability, the infinite identity and irrationality equivalence

$$\sum_{i\geq 0} \frac{p_i}{2^{i+1}} = 2 + \sum_{i\geq 0} \frac{g_i}{2^{i+1}}, \qquad \mathrm{Irr}\left(\sum_{i\geq 0} \frac{p_i}{2^{i+1}}\right) \iff \mathrm{Irr}\left(\sum_{i\geq 0} \frac{g_i}{2^{i+1}}\right).$$

The prime number theorem supplies the summability in ordinary mathematics; that external input is not formalised here and remains an explicit hypothesis of the Lean declarations.

Problem #251 remains open. The remaining obligation is now exact: rationality would force some fixed $h \geq 1$ for which the actual tail shifts $T_{N+h} - T_N$ are integers at every sufficiently large N. Thus it is enough to prove cofinal non-integrality for every fixed h. We also prove a local obstruction: two adjacent shifts lying strictly between -1 and 1 cannot both be integers when the corresponding prime gaps differ. Consequently it is enough to produce

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such adjacent small-mismatch pairs cofinally. The actual prime gaps are Lean-checked to be unbounded, hence not eventually periodic, but that fact alone gives no control on the size of a tail. A telescoping identity supplies an explicit sequence of integers $\kappa_0, \kappa_1, \ldots$ that is unbounded, hence not eventually periodic, and for which $\sum_{n\geq 0} \kappa_n 2^{-(n+1)}$ is rational. Rationality of a dyadic sum therefore cannot by itself force its integer coefficients to be eventually periodic, and the missing theorem must control the full prime-gap tails.

1 Introduction

Let p_n denote the nth prime. Erdős Problem #251 asks whether

$$\Pi = \sum_{n\geq 1} \frac{p_n}{2^n}$$

is irrational [1, p. 94][2, p. 62][4, p. 103]. Bloom's current catalogue record reproduces this question and labels it open, while explicitly warning that the status is the website owner's present assessment and may omit relevant literature [13]. We therefore use the catalogue for numbering and current reported status only; the three original publications carry the mathematical claims. The problem is open. Numerically $\Pi = 2/2 + 3/4 + 5/8 + 7/16 + 11/32 + \cdots = 3.674643966\ldots$.

Throughout the formal development the primes are indexed from zero, so that $p_0^{(0)} = 2$, $p_1^{(0)} = 3$, $p_2^{(0)} = 5$, and the gaps are $g_i = p_{i+1}^{(0)} - p_i^{(0)}$. In that indexing

$$\Pi = \sum_{i\geq 0} \frac{p_i^{(0)}}{2^{i+1}},$$

which is the convention every statement below uses; we drop the superscript from here on. The first values are

i	0	1	2	3	4	5	6	7	8	9
p_i	2	3	5	7	11	13	17	19	23	29
g_i	1	2	2	4	2	4	2	4	6	2

so $g_0 = 1$ and every later gap is even, the primes after 2 being odd. A second normalisation with denominator 2^i also occurs, and at every finite horizon it is exactly twice this one,

$$\sum_{i=0}^{n-1} \frac{p_i}{2^i} = 2\sum_{i=0}^{n-1} \frac{p_i}{2^{i+1}},$$

the factor-of-two normalisation. A factor of two does not change rationality, so no result below depends on the choice; the identity is stated so that the indexing cannot drift silently.

Notation. $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}_{>0} = \{1, 2, \ldots\}$. We write φ for Euler's totient function, $\operatorname{den} q$ for the denominator of a rational number q written in lowest terms, $\operatorname{dist}(x, \mathbb{Z})$ for the distance from a real number x to the nearest integer, and $\operatorname{Irr}(x)$ for the assertion that x is irrational. A rational number is called *integral* when it is the image of an integer. A sequence (a_n) is *eventually periodic with period* $h \geq 1$ when $a_{n+h} = a_n$ for every sufficiently large n. A sum over an empty range of indices is zero.

Relation to prior work. Erdős proved that $\sum_n p_n^k/n!$ is irrational for every $k \geq 1$ [1, p. 93]. The 1958 article gives the $k = 1$ proof on pp. 94–95 and says explicitly that the more complicated $k > 1$ proof is omitted. On page 103 of his 1988 problem paper he separately stated the fixed-denominator problem: he could not prove that $\sum_n p_n^k/2^n$ is irrational for every $k \geq 1$, and wrote that the case $k = 1$ was probably already very difficult. He also stated the variable-denominator expectation that $\sum_{n\geq 1} p_n/(g_1 \cdots g_n)$ is irrational whenever $g_n \geq 2$ and $g_n = o(p_n)$ [4, p. 103]. The latter expectation is false: ChatGPT 5.4 Pro, orchestrated by Vjeko Kovač, constructs such a sequence (g_n) for which the sum is exactly 1 [14, Theorem 1 and proof, pp. 1–2]. Under the stronger hypotheses that (g_n) is strictly increasing and $g_n = O(n \log^k n)$, Erdős had already classified the rational cases: rationality holds exactly when $g_n = qp_n + 1$ eventually for one fixed integer $q > 1$ [1, pp. 96–99]. The counterexample evades those hypotheses. It does not address the fixed-denominator dyadic series studied here. The current catalogue record still repeats this variable-denominator expectation without mentioning the counterexample [13]; its warning about possibly missing literature is therefore material here, not merely boilerplate.

There is also a genuinely adjacent proved dyadic theorem. If $P(m)$ denotes the largest prime factor of m, Erdős and Pomerance proved that

$$\sum_{m\geq 2} \frac{\mathbf{1}_{\{P(m)>P(m+1)\}}}{2^m}$$

is irrational [3, §7, p. 320]. Erdős and Graham record the complementary indicator on p. 62: equality of the two largest prime factors is impossible for consecutive integers, so its series is $1/2$ minus the displayed one and is irrational as well. This is a theorem about a bounded prime-factor comparison digit sequence, not the unbounded prime numerators in Π; it supplies nearby positive evidence without solving Problem #251.

Problem #251 also appears as the unproved declaration `erdos_251` in the *Formal Conjectures* repository [15]. Its zero-based Lean sum starts with the zeroth prime over 2^0, so it is twice the displayed normalisation and has equivalent irrationality status, but it is not literally the same indexing; its proof is `sorry`. No priority is claimed for anything below.

The strategy. The digits p_n grow, so the series is not a digit expansion in any bounded alphabet, and the standard rationality criteria for such expansions do not apply directly. The classical elementary criteria for series of this kind instead control irrationality through the growth of the denominators. Erdős and Straus named a sum $\sum_k 1/a_k$ over a strictly increasing sequence of positive integers an *Ahmes series* [5]; for such a series the condition $a_k^{1/2^k} \to \infty$ is sufficient for irrationality, and it is sharp, since shifted Sylvester sequences grow like C^{2^k} for arbitrarily large C and have rational reciprocal sum. Both statements and their attribution are recorded in the introduction of Kovač and Tao [7, §1], who develop the elementary technology for such series much further. Splitting each term $p_i/2^{i+1}$ into p_i copies of $2^{-(i+1)}$ writes Π as a sum of unit fractions, but with repetitions, and the denominators occurring in it are exactly the powers of two. Even ignoring the repetitions the growth hypothesis fails by every available margin, since $(2^n)^{1/2^n} \to 1$, and every arithmetic constraint has to come from the numerators instead.

What replaces growth control is denominator control on the sequence of rescaled tails. Rationality of the sum turns out to be equivalent to an eventual integrality condition on differences of those tails, and that condition uses nothing about the numerators beyond the fact that they are integers. The condition does not by itself force the numerators to repeat: Proposition 4.1, applied to $K_n = n$, produces the integer sequence $\kappa_n = n - 1$, which is unbounded and hence not eventually periodic, and whose dyadic sum $\sum_{n\geq 0} \kappa_n 2^{-(n+1)}$ is zero.

Outline. Section 2 replaces the primes by their consecutive gaps, the natural increments studied by prime-distribution theory. Summation by parts with the endpoint retained gives an exact finite identity, and the termwise relation $v_n = 2u_{n+1} - u_n$ between the gap terms and the prime terms makes the passage to the limit a matter of summability alone; the prime number theorem supplies that summability externally. Section 3 rescales the tails of a dyadic series into a recurrence and develops it: the block identity, the integral-shift criterion, the collapse of every rational solution onto an eventually integral shift, its contrapositive over the reals, and a local obstruction that converts an infinite-tail condition into a single comparison of gaps. Section 4 shows that the tail constraint does not by itself make the coefficient sequence eventually periodic, and Section 5 states the remaining obligation. The pinned Lean 4 toolchain and Mathlib revision check the formal statements. The cited system paper identifies Lean 4 [8, abstract and §1, pp. 625–626], while the Mathlib paper documents the library's historical Lean 3-era architecture [9, abstract and §1.1, p. 367]; it is not authority for the current pinned revision. Linked phrases open the corresponding declaration at the pinned source revision 7558c58afb7c.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Keywords. irrationality; prime gaps; dyadic series; summation by parts; Lean 4. **MSC 2020.** 11J72 (primary); 11N05, 68V20 (secondary).

2 Summation by parts, with the endpoint retained

Summation by parts trades a sequence for its consecutive differences. The form recorded here is exact: it carries no error term, it assumes nothing about the sequence, and it retains the endpoint term rather than absorbing it into an estimate. For a sequence P of rational numbers and $n \geq 0$ put

$$D(P,n) = \sum_{i=0}^{n-1} \frac{P(i)}{2^{i+1}}, \qquad \Delta(P,n) = \sum_{i=0}^{n-1} \frac{P(i+1) - P(i)}{2^{i+1}},$$

both empty, hence zero, at $n = 0$. We call these the dyadic partial sum and the dyadic difference sum of P.

Statement	Status	Treatment here
Irrationality of Π	Open	Not proved.
Finite prime-gap identity	Proved here	Theorem 2.2, with the endpoint retained.
Infinite prime-gap identity	Lean-checked conditionally	Theorem 2.3; the prime number theorem supplies the hypothesis externally.
Prime-series/gap-series irrationality equivalence	Lean-checked conditionally	Corollary 2.4.
Block identity for the tail recurrence	Proved here	Theorem 3.2.
Integral shift criterion	Proved here; an equivalence	Theorem 3.3.
Totient shift from an odd denominator	Proved here	Theorem 3.4.
Propagation of an integral shift	Proved here	Theorem 3.5.
Eventual integral shift for every rational-valued recurrence	Lean-checked	Theorem 3.6.
Cofinal shift escape implies irrationality	Lean-checked abstractly	Theorem 3.7.
Actual prime gaps are unbounded and not eventually periodic	Lean-checked	Proposition 3.10.
Adjacent small-mismatch pair excludes simultaneous integrality	Lean-checked	Theorem 3.8.
Concrete prime-gap tail recurrence	Paper-proved from PNT	Not yet instantiated as a Lean infinite sum.
Rationality alone forces periodic integer coefficients	False	Section 4, Proposition 4.1.
Cofinal adjacent small-mismatch hypothesis	Proposed sufficient theorem	Problem 5.2; not proved.

How to read the middle column. Proved here marks a statement proved in the text; each such statement also carries a link to a Lean declaration where it appears below. *Lean-checked* marks a statement the pinned kernel accepts, in the exact sense fixed by the Status paragraph above. The modifier *conditionally* marks a hypothesis that the Lean statement carries and that ordinary mathematics discharges from outside the formal development; *abstractly* marks a statement proved for an arbitrary integer digit sequence rather than for the actual prime gaps. *Paper-proved from PNT* marks a statement proved in the text from the prime number theorem and not formalised. *Proposed sufficient theorem* marks an unproved statement which, if proved, would give irrationality of Π.

Proposition 2.1 (finite summation by parts)**.** *For every rational sequence P and every $n \geq 0$,*

$$D(P, n+1) = P(0) + \Delta(P, n) - \frac{P(n)}{2^{n+1}}.$$

Proof. A routine induction on n. At $n = 0$ both sides equal $P(0)/2$. For the step, adding $P(n+1)/2^{n+2}$ to the left and $(P(n+1) - P(n))/2^{n+1}$ to the difference sum changes the endpoint term from $P(n)/2^{n+1}$ to $P(n+1)/2^{n+2}$, and the two adjustments agree. □

Formalised as the summation-by-parts identity. Nothing is assumed about P: no positivity, no monotonicity, and no convergence.

Specialising to $P(i) = p_i$, whose first value is $p_0 = 2$, and writing $g_i = p_{i+1} - p_i$ for the zero-based gaps, gives the reformulation.

Theorem 2.2 (prime-gap reformulation)**.** *Let $p_0 = 2, p_1 = 3, \ldots$ be the primes in increasing order and $g_i = p_{i+1} - p_i$. For every $n \geq 0$,*

$$\sum_{i=0}^{n} \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

Formalised as the prime-gap summation by parts, using the gap partial sum and the gap cast identity; the latter records that the natural-number difference $p_{n+1} - p_n$ agrees with the difference taken in $\mathbb{Q}$, which needs $p_n \leq p_{n+1}$, the monotonicity step. The leading 2 is the first prime, not a normalising constant. At $n = 2$, for instance, the left side is $2/2 + 3/4 + 5/8 = 19/8$ and the right side is $2 + (1/2 + 2/4) - 5/8 = 19/8$.

2.1 The infinite identity and the irrationality equivalence

Write

$$u_n = \frac{p_n}{2^{n+1}}, \qquad v_n = \frac{g_n}{2^{n+1}}$$

for the terms of the prime series and of the gap series, so that $\sum_{n\geq 0} u_n = \Pi$. The termwise identity $v_n = 2u_{n+1} - u_n$ is the dyadic discrete derivative. It expresses each gap term as an integer combination of two consecutive prime terms, so once (u_n) is summable the gap series can be summed by rearranging two copies of the prime series, which is what the following proof does.

Theorem 2.3 (infinite prime-gap identity)**.** *Let $u_n = p_n/2^{n+1}$ and $v_n = g_n/2^{n+1}$. If (u_n) is summable, then (v_n) is summable and*

$$\sum_{n\geq 0} u_n = 2 + \sum_{n\geq 0} v_n.$$

Proof. The shifted sequence (u_{n+1}) is summable. Sum $v_n = 2u_{n+1} - u_n$ and use $u_0 = 1$:

$$\sum_{n\geq 0} v_n = 2\left(\sum_{n\geq 0} u_n - u_0\right) - \sum_{n\geq 0} u_n = \sum_{n\geq 0} u_n - 2.$$

□

The summability transfer is the gap-series summability theorem, and the displayed identity is the infinite prime-gap identity.

Corollary 2.4 (exact irrationality reformulation)**.** *With u_n and v_n as in Theorem 2.3, and (u_n) summable,*

$$\operatorname{Irr}\Big(\sum_{n\geq 0} u_n\Big) \iff \operatorname{Irr}\Big(\sum_{n\geq 0} v_n\Big).$$

The corresponding zero-based series with denominator 2^n is $4 + 2\sum_{n\geq 0} v_n$ and has the same irrationality status.

These are the normalised irrationality equivalence, the displayed-series identity, and the displayed-series irrationality equivalence.

The summability hypothesis. The prime number theorem, proved by Hadamard and by de la Vallée Poussin in 1896, is proved quantitatively by Montgomery and Vaughan in Theorem 6.9 [11, pp. 179–181]. Their Exercise 6.2.5 [11, p. 183] then gives the sharper expansion

$$p_n = n\left(\log n + \log\log n - 1 + \frac{\log\log n - 2}{\log n} + O\Big(\frac{(\log\log n)^2}{(\log n)^2}\Big)\right).$$

In particular $p_n \sim n\log n$, so $p_n/2^n$ is summable by comparison with any fixed exponential ρ^n with $1 < \rho < 2$. Consequently Theorem 2.3 and Corollary 2.4 are unconditional paper-level consequences of a classical external theorem. The local Lean development does not formalise the prime number theorem, so its declarations retain summability as an explicit hypothesis. This is a gap in the formalisation, not in the mathematics: the open content is exactly irrationality of the prime-gap series. Concretely, $\Pi = 3.674643966\ldots$ is irrational if and only if $\sum_{n\geq 0} g_n 2^{-(n+1)} = 1.674643966\ldots$ is, and neither is known.

3 The tail recurrence and integral shifts

The series is not attacked directly. Suppose $\sum_{i\geq 0} a_i 2^{-(i+1)}$ converges, with every a_i an integer, and rescale its tails by putting

$$T_N = 2^{N+1}\sum_{i>N}\frac{a_i}{2^{i+1}} = \sum_{j\geq 1}\frac{a_{N+j}}{2^j}.$$

Two facts follow immediately. First $T_{N+1} = 2T_N - a_{N+1}$, so moving one level along doubles the rescaled tail and subtracts a single coefficient. Second $T_0 = 2\sum_{i\geq 0} a_i 2^{-(i+1)} - a_0$, so the sum is rational exactly when T_0 is. The whole of Section 3 therefore studies that recurrence in isolation, assuming nothing about the coefficients beyond the fact that they are integers; the prime-gap instance is resumed at the end of the section. The following definition is the object so obtained, stripped of its origin.

Definition 3.1. Let $g : \mathbb{N} \to \mathbb{Z}$ and $T : \mathbb{N} \to \mathbb{Q}$. Say T satisfies the *dyadic tail recurrence* with *digits* g when

$$T_{N+1} = 2T_N - g_{N+1} \qquad \text{for every } N.$$

We call the sequence $(T_N)_{N\geq 0}$ the *orbit* of T_0 under g, write $\sigma_h(N) = T_{N+h} - T_N$ for the *shift* of length h at N, and call a rational number *integral* when it is the image of an integer.

These are the tail recurrence, the shift, and integrality. The digits are arbitrary integers. In the intended instance they are the prime gaps and T_N is the scaled tail of Π after level N, but that instantiation needs a summability argument and is not made here; every statement below is a theorem about Definition 3.1.

One small orbit, referred to again below, is worth having in view. Take every digit $g_N = 0$ and $T_0 = 1/12$. Then $T_{N+1} = 2T_N$, so the orbit is $\frac{1}{12}, \frac{1}{6}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \ldots$, and $\sigma_h(N) = (2^h - 1)2^N/12$. The shift of length 2 is not integral at $N = 0$, where $\sigma_2(0) = \frac{1}{3} - \frac{1}{12} = \frac{1}{4}$, and is integral at $N = 2$, where $\sigma_2(2) = \frac{4}{3} - \frac{1}{3} = 1$; the shift of length 1 is $\sigma_1(N) = 2^N/12$ and is never integral. The change at $N = 2$ is accounted for by Theorem 3.6, and the failure at $h = 1$ by Theorem 3.3.

Iterating the recurrence h times multiplies T_N by 2^h and accumulates an explicit integer, which we now name. Define $B_{0,N} = 0$ and $B_{h+1,N} = 2B_{h,N} + g_{N+h+1}$, so that $B_{h,N} = g_{N+1}2^{h-1} + \cdots + g_{N+h}$: the tail block. Thus $B_{1,N} = g_{N+1}$, $B_{2,N} = 2g_{N+1} + g_{N+2}$ and $B_{3,N} = 4g_{N+1} + 2g_{N+2} + g_{N+3}$: the block puts the weights $2^{h-1}, \ldots, 2^0$ on the h digits following index N.

Theorem 3.2 (block identity)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g, and let $B_{h,N}$ be as above. For every N and h,*

$$T_{N+h} = 2^h T_N - B_{h,N}, \qquad \textit{hence} \qquad \sigma_h(N) = (2^h - 1)\,T_N - B_{h,N}.$$

Proof. A routine induction on h; the step is one application of the recurrence together with $2 \cdot 2^h = 2^{h+1}$ and the recursion defining B. The second identity is the first minus T_N. □

Formalised as the iterated block identity and the scaled shift identity. The shift also obeys the recurrence in its own right, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$, the shift step identity.

Since $B_{h,N}$ is an integer, Theorem 3.2 converts a question about the shift into a question about the single scaled term $(2^h - 1)T_N$.

Theorem 3.3 (integral-shift criterion)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g. For every N and h, the shift $\sigma_h(N)$ is integral if and only if $(2^h - 1)T_N$ is integral.*

Proof. By Theorem 3.2 the two differ by the integer $B_{h,N}$, and subtracting an integer does not change integrality. □

Formalised as the integral-shift criterion, on the integer-shift invariance. This is an equivalence, not a one-way reduction: the block carries no information about integrality, so the two conditions are the same condition. Informally, once T_N is fixed no choice of the digits after index N can change whether the shift $\sigma_h(N)$ is an integer; that is decided by the denominator of T_N alone.

Theorem 3.4 (a shift of totient length)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g, and let φ be Euler's totient function. If the reduced denominator d of T_N is odd, then $\sigma_{\varphi(d)}(N)$ is integral.*

Proof. Since d is odd, 2 and d are coprime, so Euler's congruence gives $2^{\varphi(d)} \equiv 1 \pmod{d}$, that is $d \mid 2^{\varphi(d)} - 1$. Writing $2^{\varphi(d)} - 1 = dk$ and $T_N = u/d$ in lowest terms, $(2^{\varphi(d)} - 1)T_N = ku$ is an integer, and Theorem 3.3 transfers this to the shift. □

Formalised as the totient shift and the Euler multiplier. For example, if $\operatorname{den} T_N = 3$ then $\varphi(3) = 2$ and $(2^2-1)T_N = 3T_N$ is an integer, so $\sigma_2(N)$ is integral, while $(2^1-1)T_N = T_N$ is not; if $\operatorname{den} T_N = 5$ then $\varphi(5) = 4$ and $\sigma_4(N)$ is integral.

The hypothesis is a genuine restriction: the argument uses coprimality of 2 with the denominator, and the even part of a denominator is exactly what the doubling in the recurrence acts on. It cannot be dropped. If $T_N = 1/2$ then $2^h - 1$ is odd for every $h \geq 1$, so $(2^h - 1)T_N$ is never an integer and, by Theorem 3.3, no shift at N is integral.

Theorem 3.5 (propagation)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g, and fix h and N. If $\sigma_h(N)$ is integral, then $\sigma_h(N+k)$ is integral for every $k \geq 0$.*

Proof. By the shift step identity, $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$ is an integer combination of an integer and two digits; induct on k. □

Formalised as the one-step propagation and the propagation to every later index.

The three preceding theorems combine as follows, and this is the statement the rest of the note rests on. The special case is immediate: if $\operatorname{den} T_0$ is already odd, then Theorem 3.4 at $N = 0$ makes the shift of length $\varphi(\operatorname{den} T_0)$ integral and Theorem 3.5 keeps it integral at every later index, so one may take $N_0 = 0$. In general a denominator carries a power of two as well, and the key point is that the doubling in the recurrence annihilates exactly the 2-adic part of a denominator, and nothing else: after finitely many steps the orbit therefore reaches a term with odd reduced denominator, which is precisely the situation Theorem 3.4 handles. No control of the digits is needed anywhere.

Theorem 3.6 (denominator collapse)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g (Definition 3.1). Then some fixed positive shift is integral from some point onwards:*

$$\exists h \geq 1\ \exists N_0\ \forall N \geq N_0, \qquad \sigma_h(N) \in \mathbb{Z}.$$

Proof. Write the reduced denominator of T_0 as $2^s m$ with m odd. The block identity gives $T_s = 2^s T_0 - B_{s,0}$. Multiplying by 2^s clears the entire power of two from the denominator, and subtracting the integer $B_{s,0}$ cannot reintroduce one, so the reduced denominator of T_s divides m and is in particular odd. Theorem 3.4, applied at index s, supplies the positive shift $h = \varphi(\operatorname{den} T_s)$, and Theorem 3.5 keeps that shift integral at every later index. □

In the orbit displayed after Definition 3.1 the proof runs as follows: $\operatorname{den} T_0 = 12 = 2^2 \cdot 3$, so $s = 2$, the orbit reaches $T_2 = 1/3$ with odd denominator, and $h = \varphi(3) = 2$. That is exactly the shift length seen to be integral there from index 2 onwards, and no shorter one works.

Three features of the argument are used later. The proof may take as its number of preparatory steps the 2-adic valuation of $\operatorname{den} T_0$. The resulting shift length h is the totient of the odd denominator reached from a hypothetical rational initial value, and

is not known in advance for the prime-gap orbit. This is why this argument requires Problem 5.1 for every h, rather than for one preassigned shift length. Finally, the digits enter only through the integer $B_{s,0}$, so the conclusion holds for an arbitrary integer digit sequence.
The factorisation and cancellation are the odd-denominator doubling lemma; the orbit form is the odd-denominator state theorem; and the quantified conclusion is the eventual integral-shift theorem.

The useful contrapositive is stated for a real recurrence. Call its shifts *cofinally non-integral* when, for every fixed $h \geq 1$ and every threshold N_0, some $N \geq N_0$ has $\sigma_h(N) \notin \mathbb{Z}$. This is precisely the negation of the conclusion of Theorem 3.6: no shift length whatever becomes integral and stays integral.

Theorem 3.7 (cofinal shift escape implies irrationality)**.** *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$ with integer g. If its shifts are cofinally non-integral, then T_0 is irrational.*

Proof. If T_0 were rational, the whole real orbit would be the image in $\mathbb{R}$ of the rational orbit with that initial value. Theorem 3.6 would then produce one $h \geq 1$ whose shifts are integral at every sufficiently large index, contradicting cofinal non-integrality. □

Lean checks the real recurrence and shift definitions as real dyadic recurrence and real shift, the rational-orbit transport as the rational-orbit cast theorem, cofinal non-integrality as the cofinal shift-escape predicate, and Theorem 3.7 as the cofinal-escape irrationality theorem.

The actual prime-gap orbit. Define, at paper level,

$$\mathcal{T}_N = \sum_{j\geq 1} \frac{g_{N+j}}{2^j}.$$

The prime number theorem gives convergence, and an index shift gives

$$\mathcal{T}_{N+1} = 2\mathcal{T}_N - g_{N+1}.$$

If $G = \sum_{n\geq 0} g_n/2^{n+1}$, then $\mathcal{T}_0 = 2G - g_0 = 2G - 1 = 2.349287932\ldots$. Together with Theorem 2.3, this shows that Π, G, and $\mathcal{T}_0$ have the same rationality status. Thus Theorem 3.7 applies directly to Problem #251 once cofinal shift escape is proved for $\mathcal{T}$. This last analytic instantiation is proved here from the classical prime number theorem; it is not yet a Lean declaration. The abstract theorem, including its real/rational transport, is kernel-checked.

3.1 Two adjacent small shifts cannot both be integral

Applying Theorem 3.7 requires cofinal non-integrality, which is a condition of infinite precision imposed on a complete tail. The key point is that inside the open interval $(-1, 1)$ integrality is equality with zero, so on that range the one-step shift recurrence

$$\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$$

turns simultaneous integrality of two adjacent shifts into a single comparison of digits. What this buys is a *certificate*: a condition attached to a single pair of adjacent indices, whose verification already contradicts integrality at that pair. The point of arranging the argument this way is that the distance of a shift from the integers never has to be estimated; it suffices to know that both shifts lie in $(-1, 1)$ and that two digits differ.

Theorem 3.8 (adjacent small-shift obstruction)**.** *Let $T : \mathbb{N} \to \mathbb{Q}$ satisfy the dyadic tail recurrence with integer digits g. Fix h and N. If*

$$-1 < \sigma_h(N) < 1, \qquad -1 < \sigma_h(N+1) < 1, \qquad g_{N+h+1} \neq g_{N+1},$$

then $\sigma_h(N)$ and $\sigma_h(N+1)$ cannot both be integral. Consequently, if such a pair occurs beyond every threshold, the h-shift is not eventually integral.

Proof. An integral rational strictly between -1 and 1 is zero. If both shifts were integral, both would therefore vanish, and substitution in the displayed shift step identity would give $g_{N+h+1} = g_{N+1}$, a contradiction. The cofinal statement chooses one such adjacent pair after the alleged onset of integrality. □

The finite contradiction is the adjacent small-shift obstruction; its quantified form is the cofinal small-mismatch theorem; and the actual-prime-gap specialisation is the prime-gap specialisation. Theorem 3.8 is conditional on its two tail inequalities; no theorem asserting that such pairs occur is claimed here.

All three are stated for a rational orbit, while the prime-gap tail $\mathcal{T}$ of Section 3 is real, so the rational statement does not on its own discharge the instances that arise downstream. The same argument gives the real form directly.

Corollary 3.9 (adjacent small-shift obstruction, real form)**.** *Let $T : \mathbb{N} \to \mathbb{R}$ satisfy $T_{N+1} = 2T_N - g_{N+1}$ with integer digits g, and write $\sigma_h(N) = T_{N+h} - T_N$. Fix h and N. If*

$$-1 < \sigma_h(N) < 1, \qquad -1 < \sigma_h(N+1) < 1, \qquad g_{N+h+1} \neq g_{N+1},$$

then $\sigma_h(N)$ and $\sigma_h(N+1)$ do not both lie in $\mathbb{Z}$. Consequently, if such a pair occurs beyond every threshold, the h-shift is not eventually integral.

Proof. The real recurrence gives the same shift step identity $\sigma_h(N+1) = 2\sigma_h(N) - (g_{N+h+1} - g_{N+1})$, and an integer strictly between -1 and 1 is zero. If both shifts lay in $\mathbb{Z}$, both would therefore vanish, and substitution in that identity would give $g_{N+h+1} = g_{N+1}$, a contradiction. The cofinal statement chooses one such adjacent pair after the alleged onset of integrality. □

Corollary 3.9 is proved here and is not a declaration in the pinned Lean module, whose small-shift theorems are stated for the rational orbit. It is the form that applies to the real tail, and it carries the same two unproved inequalities as its rational counterpart.

The third hypothesis, on its own, is available. For the actual gaps tabulated in Section 1 it reads $g_4 = 2 \neq 4 = g_3$ at $h = 1$, $N = 2$; and Proposition 3.10 below says precisely that for each fixed $h \geq 1$ the inequality $g_{N+h+1} \neq g_{N+1}$ holds for arbitrarily large N, since its failure from some index onwards is eventual periodicity with period h. What is missing is the pair of inequalities, each of which constrains a complete infinite tail.

Proposition 3.10 (prime gaps do not become periodic). *For every positive h, the actual consecutive-prime-gap sequence is not eventually periodic with period h.*

Proof. The gaps are unbounded, by the standard construction: the interval from $n! + 2$ to $n! + n$ contains no prime. Far stronger lower bounds for large gaps are known [6, Theorem 1, p. 66], but unboundedness is all that is needed. An eventually periodic natural-valued sequence has finite range after its preperiod, while its finite initial segment is bounded as well; hence it is bounded, a contradiction. □

Lean checks the factorial argument as unboundedness of the actual gaps and the conclusion as non-eventual periodicity. Combining nonperiodicity with eventual strict smallness of one positive shift would also exclude eventual integrality, but eventual smallness at every sufficiently large index is stronger than the cofinal adjacent-pair hypothesis in Theorem 3.8 and is not asserted here.

4 Nonperiodic coefficients with a rational sum

Proposition 3.10 shows that the prime gaps are not eventually periodic. One might therefore hope that rationality of a dyadic series forces its integer coefficients to be eventually periodic, and play the two against each other. The hoped-for implication is false, and a single explicit sequence refutes it.

The construction runs the emission of coefficients backwards. Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary, read K_n as the value carried into level n, and put $\kappa_n = 2K_n - K_{n+1}$: the carry coefficient. That definition is exactly the statement that

$$\frac{K_n}{2^n} = \frac{\kappa_n}{2^{n+1}} + \frac{K_{n+1}}{2^{n+1}},$$

so at each level the carried value splits into one emitted coefficient and a new carry. Nothing at all is assumed about K: not integrality, not positivity, not any bound. That is the sense in which the carry is free, and it is what the counterexample below exploits.

Proposition 4.1 (exact telescoping). *Let $K : \mathbb{N} \to \mathbb{Q}$ be arbitrary and $\kappa_n = 2K_n - K_{n+1}$. For every $n \geq 0$,*

$$\sum_{i=0}^{n-1} \frac{\kappa_i}{2^{i+1}} = K_0 - \frac{K_n}{2^n}.$$

Proof. A routine induction on n; the added term is $(2K_n - K_{n+1})/2^{n+1} = K_n/2^n - K_{n+1}/2^{n+1}$. □

Formalised as the carry telescoping identity, on the carry partial sum. When K takes natural-number values and $K_{n+1} \leq 2K_n$ for every n, the coefficients κ_n are natural numbers as well, and the natural-number and rational readings of the definition agree under the cast, the natural-carry cast.

Consequence for the coefficients. If $K_n 2^{-n} \to 0$, the emitted partial sums converge to K_0. For the explicit choice $K_n = n$ one has

$$\kappa_n = n - 1, \qquad \sum_{i=0}^{n-1} \frac{\kappa_i}{2^{i+1}} = -\frac{n}{2^n} \longrightarrow 0,$$

that is, the first values are

n	0	1	2	3	4	5	6
κ_n	-1	0	1	2	3	4	5
$\sum_{i<n} \frac{\kappa_i}{2^{i+1}}$	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{8}$	$-\frac{1}{4}$	$-\frac{5}{32}$	$-\frac{3}{32}$

converging to 0. The sequence (κ_n) is unbounded and therefore not eventually periodic, although its dyadic sum is rational. Thus rationality alone cannot imply eventual periodicity of an unrestricted integer coefficient sequence. Informally, an arbitrarily large value may be carried forward from level to level and still disappear in the limit, so rationality of a dyadic sum is compatible with arbitrarily large integer coefficients. This does not classify which sequences occur, and it does not show that a prescribed prime-gap pattern is emitted by a bounded or nonnegative carry; here $K_1 = 1 > 0 = 2K_0$, so this example is not of the natural-number kind above, and indeed $\kappa_0 = -1$.

5 Complements and further questions

Problem #251 is open. Two statements would close it, and neither is proved here. Both ask for indices beyond every threshold rather than for one index, because Theorem 3.6 produces integrality only from some unspecified point onwards; a single non-integral shift, however far out, is consistent with that conclusion. Throughout, $g_i = p_{i+1} - p_i$ are the zero-based prime gaps of Section 1.

Problem 5.1 (minimal prime-gap shift escape)**.** For every $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$\sum_{j\geq 1} \frac{g_{N+h+j} - g_{N+j}}{2^j} \notin \mathbb{Z}. \tag{5.1}$$

The series in (5.1) is exactly $\mathcal{T}_{N+h} - \mathcal{T}_N$. Theorem 3.7 therefore turns any solution of Problem 5.1 directly into irrationality of Π. The quantifier over h is essential: a hypothetical rational value chooses a shift length from the odd part of its reduced denominator, and that length is not known in advance.

Problem 5.2 (cofinal adjacent small mismatch)**.** For every fixed $h \geq 1$ and every N_0, prove that some $N \geq N_0$ satisfies

$$\begin{gathered} \left| \sum_{j\geq 1} \frac{g_{N+h+j} - g_{N+j}}{2^j} \right| < 1, \\ \left| \sum_{j\geq 1} \frac{g_{N+h+1+j} - g_{N+1+j}}{2^j} \right| < 1, \\ g_{N+h+1} \neq g_{N+1}. \end{gathered} \tag{5.2}$$

The two sums are adjacent h-shifts. Corollary 3.9 turns each instance of (5.2) into a finite contradiction to simultaneous integrality; Theorem 3.8 is the Lean-checked rational case. A cofinal family of such pairs therefore proves Problem 5.1; Theorem 3.7 then gives irrationality of the gap series, and Corollary 2.4 transfers it to Π. This formulation deliberately asks only for sporadic adjacent pairs; the stronger assertion that a fixed shift is eventually always smaller than one is unnecessary.

Several natural prime-distribution inputs are insufficient. Isolated small gaps, isolated large gaps, average gap estimates, and the occurrence of any one fixed finite pattern do not suffice: both inequalities in (5.2) contain the complete infinite continuation. Nor does parity: after the first gap all g_n are even. Unboundedness and non-eventual-periodicity of the actual gaps, though now checked, do not imply the required small-tail recurrence. In particular, Section 4 shows that no argument can deduce eventual periodicity from rationality alone.

A finite truncation criterion. Both problems above are stated in terms of infinite tails, but a dominated truncation reduces the first of them to a finite quantity. For $L \geq 1$ put

$$S_{h,N,L} = \sum_{j=1}^{L} \frac{g_{N+h+j} - g_{N+j}}{2^j},$$

the truncation of the series in (5.1) after L terms. This is a finite sum of gap differences and can be computed; the point of the following proposition is to say how far from an integer it must be before the discarded tail is irrelevant.

Proposition 5.3 (finite truncation). *Let $M(n) \geq g_n$ for every n, assume that the series below converges, and put*

$$R_{h,N,L}(M) = \sum_{j>L} \frac{M(N+h+j) + M(N+j)}{2^j}.$$

Suppose that for every fixed $h \geq 1$ and every N_0 there exist $N \geq N_0$ and $L \geq 1$ with

$$\operatorname{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M). \tag{5.3}$$

Then Problem 5.1 holds.

Proof. The part of $\sum_{j\geq 1}(g_{N+h+j} - g_{N+j})2^{-j}$ omitted from $S_{h,N,L}$ has absolute value at most $R_{h,N,L}(M)$, so under (5.3) the full sum lies at positive distance from every integer. □

The last distance-to-integers inference is Lean-checked as the generic finite-approximation certificate: an error at most R cannot reach an integer when the approximation is farther than R from every integer. For instance, if $S_{h,N,L} = 2/5$ and $R_{h,N,L}(M) = 1/20$, the full sum lies within $1/20$ of $2/5$ and so at distance at least $7/20$ from every integer, which settles (5.1) at that N. The prime-gap tail bound, the convergence used above, and the existence of blocks satisfying (5.3) are not consequences of that formal theorem.

Taking the classical bound $M(n) \ll n \log n$, a choice $L = \lceil A \log_2(N + h + 2) \rceil$ with any fixed $A > 1$ makes the right side a negative power of N up to logarithms. Thus (5.3) asks for a finite dyadic anti-concentration estimate on a logarithmic-length block, not control of an infinite tail and not eventual periodicity of the full gap sequence. For Problem 5.2, the same truncation must certify two adjacent full-tail values inside the open unit interval, together with the displayed gap mismatch. A finite prefix is useful only when its omitted tail is rigorously dominated.

What (5.3) requires is joint control of the finite block of weighted differences $(g_{N+h+1} - g_{N+1}, \cdots, g_{N+h+L} - g_{N+L})$ modulo powers of two, along a block of logarithmic length. We do not know how to obtain such control and we make no progress on it here. The strongest results on prime gaps address a different shape of question. Zhang's bounded-gap theorem [12, Theorem 1, p. 1122] produces infinitely many bounded consecutive-prime gaps. Maynard proves substantially more than an individual-gap statement: [10, Theorem 1.1, p. 384] bounds $\liminf_n (p_{n+m} - p_n)$ for every fixed m, and hence gives bounded clusters of every fixed size; [10, Theorem 1.3, p. 385] gives the explicit unconditional bound $\liminf_n (p_{n+1} - p_n) \leq 600$. The large-gap theorem of Ford, Green, Konyagin, Maynard and Tao [6, Theorem 1, p. 66] bounds the largest single consecutive-prime gap below X. None of these results supplies the joint dyadic distribution of a logarithmic block of consecutive gap differences required by (5.3). The same introduction notes a separate sequel on chains of large gaps; the cited theorem itself supplies no residue-sensitive block estimate of the kind needed here.

What remains to be formalised. The abstract irrationality theorem and the prime-specific local small-mismatch theorem are Lean-checked, as is the generic finite-approximation certificate used in Proposition 5.3. The local source still takes summability of the prime series as a hypothesis and does not define $\mathcal{T}_N$ by an infinite sum; the prime-number-theorem comparison and the identification of that concrete tail with the checked real recurrence are paper proofs, as is the real form of the small-shift obstruction in Corollary 3.9. The local source's prime-gap tail domination and anti-concentration hypotheses also remain paper-level obligations. No theorem supplies the cofinal pairs in (5.2), and none of the prime-distribution estimates described above is claimed or formalised.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 7558c58afb7c. All declarations of this note live in one module. The summation-by-parts declarations are prime-specific; most of Section 3 is stated for arbitrary integer digits and arbitrary rational or real orbits, while Section 3.1 records the actual-gap specialisation. The concrete prime-gap tail and its convergence are paper-level consequences of the prime number theorem; they are not definitions in the pinned Lean module.

References

[1] P. Erdős, *Sur certaines séries à valeur irrationnelle*, Enseign. Math. (2) **4** (1958), 93–100, doi:10.5169/seals-34629. The dyadic prime series is stated as unproved on p. 94; the factorial-prime family is stated on p. 93, with only the $k = 1$ proof printed on pp. 94–95.

[2] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 62.

[3] P. Erdős and C. Pomerance, *On the largest prime factors of n and $n + 1$*, Aequationes Math. **17** (1978), 311–321, doi:10.1007/BF01818569. The unnumbered dyadic irrationality theorem and its complete proof are in §7 on p. 320.

[4] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[5] P. Erdős and E. G. Straus, *On the irrationality of certain Ahmes series*, J. Indian Math. Soc. (N.S.) **27** (1964), 129–133. MR 175848.

[6] K. Ford, B. Green, S. Konyagin, J. Maynard and T. Tao, *Long gaps between primes*, J. Amer. Math. Soc. **31** (2018), 65–105, doi:10.1090/jams/876. Theorem 1 on p. 66 gives the effective lower bound for the largest single consecutive-prime gap below X.

[7] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. **175** (2025), 572–608.

[8] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in A. Platzer and G. Sutcliffe (eds.), CADE 28, Lecture Notes in Comput. Sci. 12699, Springer, 2021, pp. 625–635, doi:10.1007/978-3-030-79876-5_37.

[9] The mathlib Community, *The Lean mathematical library*, in CPP 2020, ACM, 2020, pp. 367–381, doi:10.1145/3372885.3373824. The article describes a December 2019 Lean 3-era snapshot; the repository lock owns the current revision.

[10] J. Maynard, *Small gaps between primes*, Ann. of Math. (2) **181** (2015), 383–413.

[11] H. L. Montgomery and R. C. Vaughan, *Multiplicative Number Theory I: Classical Theory*, Cambridge Stud. Adv. Math. 97, Cambridge UP, 2007, Chapter 6, pp. 168–198; Theorem 6.9, pp. 179–181, and Exercise 6.2.5, p. 183, doi:10.1017/CBO9780511618314.008.

[12] Y. Zhang, *Bounded gaps between primes*, Ann. of Math. (2) **179** (2014), 1121–1174.

[13] T. F. Bloom, *Erdős Problem #251*, `erdosproblems.com/251`, accessed 28 July 2026 (page displays "last edited 28 September 2025"). The current record labels the main dyadic problem open, cites `[Er58b]`, `[ErGr80, p. 62]` and `[Er88c, p. 103]`, and explicitly describes its status as the website owner's present assessment rather than a literature-completeness guarantee; it does not mention the 2026 counterexample in [14] to the adjacent variable-denominator conjecture.

[14] ChatGPT 5.4 Pro (orchestrated by V. Kovač), *On the Erdős problem #251*, unpublished note, 2026, hosted by the Department of Mathematics, University of Zagreb, `web.math.pmf.unizg.hr`, accessed 28 July 2026.

[15] The Formal Conjectures Authors, *FormalConjectures.ErdosProblems.*`251`, Lean source at commit `f776d2f`, 2025, accessed 28 July 2026.